

Errata

A Reactor Physics Framework to Detect Anomalies in HTGR Core

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This document lists corrections to the dissertation entitled “*A Reactor Physics Framework to Detect Anomalies in HTGR Core*”. The errors do not affect the main results or conclusions unless otherwise stated.

Detailed Corrections

- Page 10, Equation (2.13): Replace the following equation

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \delta \phi_g(\mathbf{r}, t)}{\partial t} - \nabla D_g(\mathbf{r}) \nabla \delta \phi_g(\mathbf{r}, t) + \left(\Sigma_{t,g,0}(\mathbf{r}) \delta \phi_g(\mathbf{r}, t) + \delta \Sigma_{t,g}(\mathbf{r}, t) \phi_{g,0}(\mathbf{r}) \right) = \\ \sum_{g'=1}^G \left(\Sigma_{s0,g',g,0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \delta \Sigma_{s0,g',g}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right) \\ \chi_g^p \frac{1 - \beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \chi_g \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right) + \lambda \delta C(\mathbf{r}, t) \end{aligned}$$

with the corrected version:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \delta \phi_g(\mathbf{r}, t)}{\partial t} - \nabla D_g(\mathbf{r}) \nabla \delta \phi_g(\mathbf{r}, t) + \left(\Sigma_{t,g,0}(\mathbf{r}) \delta \phi_g(\mathbf{r}, t) + \delta \Sigma_{t,g}(\mathbf{r}, t) \phi_{g,0}(\mathbf{r}) \right) = \\ \sum_{g'=1}^G \left(\Sigma_{s0,g',g,0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \delta \Sigma_{s0,g',g}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right) \\ \chi_g^p \frac{1 - \beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \chi_g \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right) + \chi_g^d \lambda \delta C(\mathbf{r}, t) \end{aligned}$$

- Page 10, Equation (2.14): Replace the following equation

$$\frac{\partial \delta C(\mathbf{r}, t)}{\partial t} = -\lambda \delta C(\mathbf{r}, t) + \chi_g^d \frac{\beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right)$$

with the corrected version:

$$\frac{\partial \delta C(\mathbf{r}, t)}{\partial t} = -\lambda \delta C(\mathbf{r}, t) + \frac{\beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, t) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, t) \phi_{g',0}(\mathbf{r}) \right)$$

- Page 11, Equation (2.15): Replace the following equation

$$\begin{aligned} \frac{i\omega}{v_g} \delta \phi_g(\mathbf{r}, \omega) - \nabla D_g(\mathbf{r}) \nabla \delta \phi_g(\mathbf{r}, \omega) + \left(\Sigma_{t,g,0}(\mathbf{r}) \delta \phi_g(\mathbf{r}, \omega) + \delta \Sigma_{t,g}(\mathbf{r}, \omega) \phi_{g,0}(\mathbf{r}) \right) = \\ \sum_{g'=1}^G \left(\Sigma_{s0,g',g,0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \delta \Sigma_{s0,g',g}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right) \\ \chi_g^p \frac{1 - \beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right) + \lambda \delta C(\mathbf{r}, \omega) \end{aligned}$$

with the corrected version:

$$\begin{aligned} \frac{i\omega}{v_g} \delta \phi_g(\mathbf{r}, \omega) - \nabla D_g(\mathbf{r}) \nabla \delta \phi_g(\mathbf{r}, \omega) + \left(\Sigma_{t,g,0}(\mathbf{r}) \delta \phi_g(\mathbf{r}, \omega) + \delta \Sigma_{t,g}(\mathbf{r}, \omega) \phi_{g,0}(\mathbf{r}) \right) = \\ \sum_{g'=1}^G \left(\Sigma_{s0,g',g,0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \delta \Sigma_{s0,g',g}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right) \\ \chi_g^p \frac{1 - \beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right) + \chi_g^d \lambda \delta C(\mathbf{r}, \omega) \end{aligned}$$

- Page 11, Equation (2.16): Replace the following equation

$$i\omega \delta C(\mathbf{r}, \omega) = -\lambda \delta C(\mathbf{r}, \omega) + \chi_g^d \frac{\beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right)$$

with the corrected version:

$$i\omega \delta C(\mathbf{r}, \omega) = -\lambda \delta C(\mathbf{r}, \omega) + \frac{\beta_{\text{eff}}}{k_{\text{eff}}} \left(\sum_{g'=1}^G \nu \Sigma_{f,g',0}(\mathbf{r}) \delta \phi_{g'}(\mathbf{r}, \omega) + \sum_{g'=1}^G \delta \nu \Sigma_{f,g'}(\mathbf{r}, \omega) \phi_{g',0}(\mathbf{r}) \right)$$